



Arithmetic sequences

Task A: Identifying arithmetic sequences

- 1) For each set of terms, work out if they could form an **arithmetic sequence** or not. Explain your answer.

a) $\boxed{5}$, $\boxed{9}$, $\boxed{13}$, $\boxed{17}$, $\boxed{21}$, $\boxed{25}$, ...

Yes. Add 4 each time

b) $\boxed{0.1}$, $\boxed{0.15}$, $\boxed{0.2}$, $\boxed{0.25}$, $\boxed{0.3}$, $\boxed{0.35}$, ...

Yes. Add 0.05 each time

c) $\boxed{4}$, $\boxed{}$, $\boxed{16}$, $\boxed{}$, $\boxed{}$, $\boxed{28}$, ...

No. For $T_1=4$ and $T_3=16$ the rule is add 6. But that would make $T_5=28$ not T_6

d) $\boxed{8}$, $\boxed{6}$, $\boxed{4}$, $\boxed{2}$, $\boxed{0}$, $\boxed{-2}$, ...

Yes. Add -2 each time

e) $\boxed{2\frac{1}{2}}$, $\boxed{}$, $\boxed{}$, $\boxed{2}$, $\boxed{}$, $\boxed{\frac{5}{3}}$, ...

$$2\frac{1}{2} = \frac{15}{6}$$

$$2 = \frac{12}{6}$$

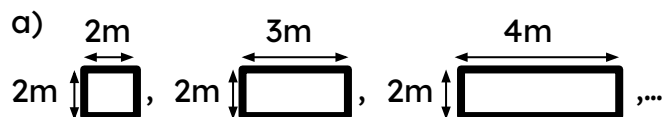
$$\frac{5}{3} = \frac{10}{6}$$

$\boxed{\frac{15}{6}}$, $\boxed{\frac{14}{6}}$, $\boxed{\frac{13}{6}}$, $\boxed{\frac{12}{6}}$, $\boxed{\frac{11}{6}}$, $\boxed{\frac{10}{6}}$, ...

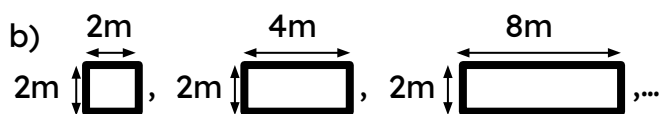
Yes. Add $-\frac{1}{6}$ each time

2) For each sequence work out the perimeter and area of each term and state whether they form a **linear sequence** or not.

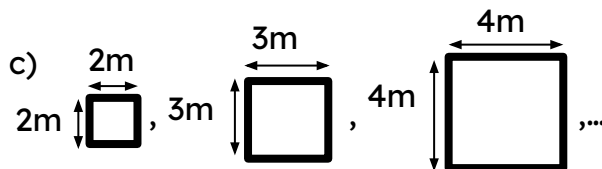
Linear sequences



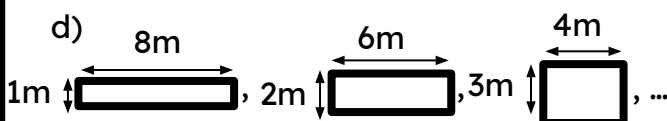
Term	1	2	3
Perimeter	8	10	12
Area	4	6	8



Term	1	2	3
Perimeter	8	12	20
Area	4	8	16



Term	1	2	3
Perimeter	8	12	16
Area	4	9	16



Term	1	2	3
Perimeter	18	16	14
Area	8	12	12

Task B: Describing arithmetic sequences

1) For each sequence state whether it is **arithmetic** or not. If it is **arithmetic**, describe the sequence.

a) 6, 15, 24, 33, ...

Arithmetic- initial term 6 and common difference 9

b) 5, 12, 15, 22, ...

Not arithmetic- difference of 7 then 3

c) 5, -3, -11, -19, ...

Arithmetic- Initial term 5 and common difference -8

d) 1, 3, 9, 27, ...

Not arithmetic- difference of 2, then 6, then 18

e) 1.5, 2.6, 3.7, 5.8, ...

Not arithmetic- differences of 1.1, 1.1 and 2.1

f) 25, 15, 10, 5, ...

Not arithmetic- difference of -10, then -5, then -5

g) $x, x + 2, x + 4, x + 6, \dots$

Arithmetic- initial term x and common difference of 2

h) $2x, 4x, 8x, 16x, \dots$

Not arithmetic- difference of $2x, 4x, 8x$

i) $x, 2x + 1, 3x + 2, 4x + 3, \dots$

Arithmetic- initial term x and common difference of $x + 1$

j) $x - 2, x, x + 2, x + 4, \dots$

Arithmetic- initial term $x - 2$ and common difference of 2

2) Below are the first four terms in four different **linear sequences**: Fill in the table with true or false for each sequence.

Sequence	1, 5, 9, 13,..	7, 13, 19, 25,..	80, 75, 70, 65,..	22, 19, 16, 13,..
Contains negative numbers	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>
Common difference is positive	<i>True</i>	<i>True</i>	<i>False</i>	<i>False</i>
Has 1 as a term	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>
Has 25 as a term	<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>
Has values greater than 50	<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>